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## Appendix G — Physical and Mathematical Formulation of the Model and FNRR

### Purpose

This appendix presents the physical and mathematical formulation supporting the doctoral thesis:

**"Estudio de la velocidad de viento e inclusión de parámetros físicos para la predicción de energía eléctrica producida por fuentes eólicas".**

Its purpose is to formalize the physical construction of Wind Power Density, the predictive structure of the model, the uncertainty representation through PI90 intervals, the energy-integration logic, and the definition of the **Factor de No Regularidad Regional (FNRR)**.

This appendix supports the scientific connection between:

meteorological observations
→ physical variables
→ Wind Power Density
→ predictive modeling
→ uncertainty quantification
→ FNRR
→ free energy
→ usable energy
→ regional energy projection

The appendix is written as a mathematical support document for repository review, thesis defense, and future derivative products.

## G.1 General structure of the physical–statistical system

The doctoral model can be represented as a structured physical–statistical system:

$S = \{D, P, \Gamma, M, U, \Psi, \Omega\}$

where:

| Symbol | Meaning |
|---|---|
| $D$ | Observational meteorological dataset. |
| $P$ | Preprocessing and traceability operator. |
| $\Gamma$ | Physical transformation from meteorological variables to energetic variables. |
| $M$ | Predictive modeling operator. |
| $U$ | Uncertainty quantification operator. |
| $\Psi$ | Structural irregularity operator associated with FNRR. |
| $\Omega$ | Energy integration and projection operator. |

The complete workflow can be expressed as:

$D_raw \rightarrow P(D_raw) \rightarrow \Gamma(P(D_raw)) \rightarrow M(\Gamma) \rightarrow U(M) \rightarrow \Psi(WPD) \rightarrow \Omega(E_free, FNRR)$

This expression summarizes the central logic of the thesis: the final energetic interpretation is not obtained directly from raw data, but from a sequence of physically and computationally traceable transformations.

## G.2 Observational state vector

For each analytical zone z and time index t, the meteorological state may be represented as:

$x_z(t) = [v_z(t), d_z(t), p_z(t), T_z(t), RH_z(t), PR_z(t), EV_z(t), NU_z(t), FA_z(t)]$

where:

| Symbol | Meaning |
|---|---|
| $v_z(t)$ | Wind speed in zone z at time t. |
| $d_z(t)$ | Wind direction. |
| $p_z(t)$ | Atmospheric pressure. |
| $T_z(t)$ | Air temperature. |
| $RH_z(t)$ | Relative humidity. |
| $PR_z(t)$ | Precipitation. |
| $EV_z(t)$ | Evaporation. |

| Symbol | Meaning |
|---|---|
| $NU_z(t)$ | Cloudiness. |
| $FA_z(t)$ | Atmospheric phenomenon or qualitative atmospheric descriptor. |

The primary observed variable for energetic modeling is wind speed:

$v_z(t) \geq 0$

The derived physical target variable is Wind Power Density:

$WPD_z(t)$

The model therefore moves from an atmospheric state vector to a physical energetic quantity.

## G.3 Physical derivation of Wind Power Density

Wind power density is derived from the kinetic energy flux of moving air.

The kinetic energy per unit mass of an air parcel moving with speed v is:

$e_k = 0.5 \cdot v^2$

The mass flow rate per unit area crossing a surface perpendicular to the wind direction is:

$\dot{m}_A = \rho \cdot v$

where:

| Symbol | Meaning |
|---|---|
| $\rho$ | Air density. |
| $v$ | Wind speed. |
| $\dot{m}_A$ | Mass flow rate per unit area. |

The kinetic power per unit area is obtained by multiplying kinetic energy per unit mass by mass flow rate per unit area:

$WPD = e_k \cdot \dot{m}_A$

Substituting:

$WPD = (0.5 \cdot v^2)(\rho \cdot v)$

Therefore:

$WPD = 0.5 \cdot \rho \cdot v^3$

This is the central physical equation of the thesis.

## G.4 Dimensional consistency of WPD

The units are:

$\rho \rightarrow kg/m^3$
$v \rightarrow m/s$
$v^3 \rightarrow m^3/s^3$

Then:

$\rho \cdot v^3 = (kg/m^3)(m^3/s^3) = kg/s^3$

Since:

$1\ W = 1\ kg \cdot m^2/s^3$

then:

$1\ W/m^2 = 1\ kg/s^3$

Therefore:

$WPD = 0.5 \cdot \rho \cdot v^3$

has units of:

$W/m^2$

This confirms that the expression is dimensionally consistent.

## G.5 Air-density approximation

Air density can be estimated using the ideal gas approximation:

$\rho = p / (R_d \cdot T_K)$

where:

| Symbol | Meaning |
|---|---|
| $\rho$ | Air density in $kg/m^3$. |
| p | Atmospheric pressure in Pa. |
| R_d | Specific gas constant for dry air. |
| T_K | Absolute temperature in K. |

The absolute temperature is:

$T_K = T_C + 273.15$

where T_C is temperature in degrees Celsius.

The dry-air constant is:

$R_d \approx 287.05\ J \cdot kg^{-1} \cdot K^{-1}$

Thus, the operational physical chain is:

$p, T \rightarrow \rho$
$v, \rho \rightarrow WPD$

If humidity correction is explicitly implemented, the moist-air density can be expressed as:

$\rho = (p_d / (R_d \cdot T_K)) + (e / (R_v \cdot T_K))$

where:

| Symbol | Meaning |
|---|---|
| $p_d$ | Partial pressure of dry air. |
| $e$ | Water-vapor pressure. |
| $R_v$ | Specific gas constant for water vapor. |

However, unless the code explicitly implements the moist-air correction, the operational thesis formulation should be interpreted through the dry-air approximation or the specific implemented density formula.

## G.6 Nonlinear sensitivity of WPD to wind speed

Because:

$WPD = 0.5 \cdot \rho \cdot v^3$

the derivative of WPD with respect to wind speed is:

$\partial WPD/\partial v = 1.5 \cdot \rho \cdot v^2$

This derivative shows that sensitivity increases quadratically with wind speed.

The relative differential form is:

$dWPD / WPD = d\rho / \rho + 3 \cdot dv / v$

This means that a small relative change in wind speed produces approximately three times that relative change in WPD, assuming air density remains locally constant.

Therefore:

wind-speed variability → amplified energetic variability

This justifies why the thesis does not treat wind speed only as a meteorological variable, but as a physically amplified energetic driver.

## G.7 Zonal formulation

Let:

$z \in \{1, 2, 3, 4\}$

represent the four analytical zones.

For each zone z, the wind-speed series is:

$V_z = \{v_z(t_1), v_z(t_2), \ldots, v_z(t_n)\}$

The corresponding WPD series is:

$Y_z = \{WPD_z(t_1), WPD_z(t_2), \ldots, WPD_z(t_n)\}$

where:

$WPD_z(t_i) = 0.5 \cdot \rho_z(t_i) \cdot v_z(t_i)^3$

The zonal structure allows the thesis to interpret the atmospheric system as a set of regional subsystems rather than as a single homogeneous domain.

## G.8 Physical interpretation of the zone

A zone is not treated as a purely administrative unit.

In the thesis, a zone is an operational atmospheric subsystem:

$Z_z = \{\text{stations, variables, temporal records, derived physical quantities}\}$

The zone supports:

- regional physical characterization,
- predictive modeling,
- uncertainty interpretation,
- FNRR computation,
- free-energy estimation,
- usable-energy estimation,
- and projection toward 2028.

Thus, the zone is the spatial unit through which the atmospheric signal becomes physically and computationally interpretable.

## G.9 Predictive formulation

Let h represent the prediction horizon.

The target variable may be represented as:

$Y_z(t+h) = WPD_z(t+h)$

A predictive model m estimates:

$\hat{Y}_{z,m}(t+h) = f_m(\Phi_z(t); \theta_m)$

where:

| Symbol | Meaning |
|---|---|
| $\hat{Y}_{z,m}(t+h)$ | Predicted WPD for zone z, model m, and horizon h. |
| $f_m$ | Predictive function associated with model m. |
| $\Phi_z(t)$ | Feature vector or lagged information available at time t. |
| $\theta_m$ | Parameters or hyperparameters of model m. |

The feature vector may include:

$\Phi_z(t) = [Y_z(t), Y_z(t-1), \ldots,$ meteorological covariates, temporal indicators$]$

depending on the model family and implementation.

## G.10 Model families

The predictive operator may include several model families:

$M = \{M_{classical}, M_{ML}, M_{DL}, M_{hybrid}\}$

where:

| Model family | Interpretation |
|---|---|
| $M_{classical}$ | Classical time-series models such as ARIMA or ARIMAX. |
| $M_{ML}$ | Machine learning models such as Random Forest or XGBoost. |
| $M_{DL}$ | Deep learning models such as LSTM. |
| $M_{hybrid}$ | Hybrid physical–statistical integration stage. |

The purpose of using several model families is not only to compare algorithms, but to evaluate which modeling structure captures the physical and temporal behavior of the wind-energy signal more effectively.

## G.11 Persistence benchmark

The persistence benchmark is defined as a physically meaningful reference model.

For horizon h, a simple persistence prediction can be expressed as:

$\hat{Y}_{z,pers}(t+h) = Y_z(t)$

This assumes that the future value is approximated by the most recent available value.

Persistence is relevant because atmospheric variables often contain temporal dependence.

A model is scientifically useful only if it improves over this baseline.

## G.12 Deterministic evaluation metrics

For observed values $Y_i$ and predictions $\hat{Y}_i$, the residual is:

$e_i = Y_i - \hat{Y}_i$

The Root Mean Squared Error is:

$RMSE = \mathrm{sqrt}((1/n) \cdot \Sigma(e_i^2))$

The Mean Absolute Error is:

$MAE = (1/n) \cdot \Sigma |e_i|$

The coefficient of determination is:

$R^2 = 1 - [\Sigma(Y_i - \hat{Y}_i)^2 / \Sigma(Y_i - \bar{Y})^2]$

where:

$\bar{Y} = (1/n) \cdot \Sigma Y_i$

These metrics provide deterministic information about predictive error and explanatory capacity.

## G.13 Skill Score versus persistence

The Skill Score evaluates improvement relative to persistence.

Using RMSE as the loss measure:

$Skill = 1 - (RMSE_{model} / RMSE_{persistence})$

Interpretation:

$Skill > 0 \rightarrow$ the model improves over persistence
$Skill = 0 \rightarrow$ the model performs similarly to persistence
$Skill < 0 \rightarrow$ the model performs worse than persistence

A stronger condition may be expressed as:

$$\text{Skill} > 0.50$$

which indicates that the model reduces predictive error by more than 50% relative to persistence, under this RMSE-based definition.

This criterion is central because it prevents the thesis from selecting a model only because it produces acceptable isolated metrics. The model must demonstrate improvement over a physically meaningful temporal baseline.

## G.14 Prediction residuals

The residual series is:

$$R_{z,m,h} = \{e_{z,m,h}(t_i)\}$$

where:

$$e_{z,m,h}(t_i) = Y_z(t_{i+h}) - \hat{Y}_{z,m}(t_{i+h})$$

Residual analysis is required because global metrics may hide systematic error behavior.

Residuals should be inspected in relation to:

- zone,
- horizon,
- target variable,
- model family,
- and temporal structure.

A model with low RMSE but structured residuals may still require methodological caution.

## G.15 PI90 uncertainty interval

A 90% prediction interval for model m, zone z, and horizon h can be expressed as:

$$PI90_{z,m,h}(t) = [L_{z,m,h}(t), U_{z,m,h}(t)]$$

where:

| Symbol | Meaning |
|---|---|
| $L_{z,m,h}(t)$ | Lower bound of the prediction interval. |
| $U_{z,m,h}(t)$ | Upper bound of the prediction interval. |

A generic asymmetric residual-based interval may be written as:

$$L_{z,m,h}(t) = \hat{Y}_{z,m}(t+h) + c_{PI90} \cdot q_{0.05}(R_{z,m,h})$$

$$U_{z,m,h}(t) = \hat{Y}_{z,m}(t+h) + c_{PI90} \cdot q_{0.95}(R_{z,m,h})$$

where:

| Symbol | Meaning |
|---|---|
| $q_{0.05}(R)$ | 5th percentile of the residual distribution. |
| $q_{0.95}(R)$ | 95th percentile of the residual distribution. |
| $c_{PI90}$ | Calibration factor. |

A generic symmetric absolute-residual interval may alternatively be expressed as:

$$L_{z,m,h}(t) = \hat{Y}_{z,m}(t+h) - c_{PI90} \cdot q_{0.90}(|R_{z,m,h}|)$$

$$U_{z,m,h}(t) = \hat{Y}_{z,m}(t+h) + c_{PI90} \cdot q_{0.90}(|R_{z,m,h}|)$$

The exact interval construction should follow the implemented code, but the mathematical role is the same: to provide an uncertainty range around the central prediction.

## G.16 Empirical coverage of PI90

The empirical coverage of PI90 is:

$$Coverage_{PI90} = (1/n) \cdot \Sigma\, I[L_i \leq Y_i \leq U_i]$$

where:

| Symbol | Meaning |
|---|---|
| $I$ | Indicator function. |
| $Y_i$ | Observed value. |
| $L_i$ | Lower interval bound. |
| $U_i$ | Upper interval bound. |

A calibrated 90% interval should satisfy:

$$Coverage_{PI90} \approx 0.90$$

This does not mean that every individual observation will fall inside the interval.

It means that, under the validation conditions, approximately 90% of observations should be contained within the interval.

## G.17 Calibration factor

The calibration factor $c_{PI90}$ is introduced to adjust interval width.

A formal calibration objective can be written as:

$c^*_{PI90} = \arg\min_c \, |\mathrm{Coverage}_{PI90}(c) - 0.90|$

The purpose of calibration is to align empirical coverage with nominal coverage.

If intervals are too narrow, empirical coverage will be below 0.90.

If intervals are too wide, empirical coverage may exceed 0.90 but with low sharpness.

Thus, calibration balances reliability and interval width.

## G.18 Energy integration over a horizon

Let $\Delta t$ be the temporal resolution in hours.

For hourly data:

$\Delta t = 1$ h

The horizon-integrated energy density can be expressed as:

$$E_{h,z}(t) = (\Delta t / 1000) \cdot \Sigma_{i=1}^{h} WPD_z(t+i)$$

where:

| Symbol | Meaning |
|---|---|
| $E_{h,z}(t)$ | Horizon-integrated energy density for zone z. |
| h | Prediction horizon. |
| $WPD_z(t+i)$ | Wind Power Density at future step i. |
| 1000 | Conversion from Wh/m² to kWh/m² when WPD is in W/m² and time is in hours. |

If predicted values are used:

$$\hat{E}_{h,z}(t) = (\Delta t / 1000) \cdot \Sigma_{i=1}^{h} \hat{W}PD_z(t+i)$$

This connects prediction with energetic interpretation.

## G.19 Annual free energy

For a time window T, the free energy density is defined as:

$$E_{free,z}(T) = (\Delta t / 1000) \cdot \Sigma_{t \in T} WPD_z(t)$$

If projection outputs are used:

$$\hat{E}_{free,z}(T) = (\Delta t / 1000) \cdot \Sigma_{t \in T} \hat{W}PD_z(t)$$

This represents the physically available energetic potential before structural irregularity adjustment.

Free energy is therefore the gross energy implied by the WPD behavior.

## G.20 Need for a structural irregularity descriptor

A region may present high WPD or high free energy, but this does not imply that the energy is structurally regular or usable.

Wind-energy interpretation requires distinguishing between:

available energetic magnitude

and

structurally usable energetic behavior

The thesis introduces FNRR to represent regional irregularity in a compact, dimensionless, robust form.

## G.21 Formal definition of FNRR

Let:

$Y_z(T) = \{WPD_z(t) : t \in T\}$

be the Wind Power Density sample for zone z over a temporal window T.

Let:

$Q_{25,z}(T)$ = 25th percentile of $Y_z(T)$
$Q_{75,z}(T)$ = 75th percentile of $Y_z(T)$

where both quantiles are expressed in the same unit as WPD, typically:

$W/m^2$

The interquartile range is:

$IQR_z(T) = Q_{75,z}(T) - Q_{25,z}(T)$

The **Factor de No Regularidad Regional (FNRR)** is defined as:

$FNRR_z(T) = IQR_z(T) / [Q_{75,z}(T) + Q_{25,z}(T) + \varepsilon_z]$

where:

| Symbol | Meaning | Unit |
|---|---|---|
| $FNRR_z(T)$ | Regional non-regularity factor for zone z and temporal window T. | Dimensionless |
| $IQR_z(T)$ | Interquartile range of WPD. | $W/m^2$ |

| Symbol | Meaning | Unit |
|---|---|---|
| $Q_{75,z}(T)$ | Upper quartile of WPD. | $W/m^2$ |
| $Q_{25,z}(T)$ | Lower quartile of WPD. | $W/m^2$ |
| $\varepsilon_z$ | Small positive numerical-stability term. | $W/m^2$ |

The term $\varepsilon_z$ must have the same physical unit as WPD in order to preserve dimensional consistency.

A practical choice is:

$$\varepsilon_z > 0$$

with $\varepsilon_z$ sufficiently small relative to the characteristic WPD scale of the corresponding zone.

This definition is robust because it is based on quantiles rather than extreme values.

The numerator measures robust central dispersion, while the denominator normalizes that dispersion with respect to the robust energetic level of the zone.

Therefore, FNRR is interpreted as a dimensionless structural descriptor of regional irregularity in the WPD distribution.

## G.22 Mathematical properties of FNRR

Assume that Wind Power Density is nonnegative:

$$WPD_z(t) \geq 0$$

Then the quartiles satisfy:

$$Q_{75,z}(T) \geq Q_{25,z}(T) \geq 0$$

Therefore:

$$IQR_z(T) = Q_{75,z}(T) - Q_{25,z}(T) \geq 0$$

Since:

$$Q_{75,z}(T) + Q_{25,z}(T) + \varepsilon_z > 0$$

for $\varepsilon_z > 0$, the denominator is strictly positive.

Thus:

$$FNRR_z(T) \geq 0$$

Now, because:

$$Q_{75,z}(T) - Q_{25,z}(T) \leq Q_{75,z}(T) + Q_{25,z}(T)$$

and:

$Q_{75,z}(T) + Q_{25,z}(T) < Q_{75,z}(T) + Q_{25,z}(T) + \varepsilon_z$

it follows that:

$FNRR_z(T) < 1$

for $\varepsilon_z > 0$.

Therefore, the strict mathematical bound is:

$0 \leq FNRR_z(T) < 1$

A conservative practical statement is:

$0 \leq FNRR_z(T) \leq 1$

This proves that FNRR is nonnegative, bounded, and dimensionless.

The descriptor is mathematically consistent because the numerator and denominator have the same physical unit, and their quotient is dimensionless.

## G.23 Interpretation of FNRR values

The interpretation is:

$FNRR \approx 0$

indicates low interquartile irregularity.

$FNRR \rightarrow 1$

indicates high structural irregularity relative to the robust central range of WPD.

Therefore, FNRR is interpreted as a regional structural descriptor.

It is not a universal physical constant.

It depends on:

- the zone,
- the temporal window,
- the WPD construction,
- the preprocessing criteria,
- and the quantile definition used in the implementation.

## G.24 Why quantiles are used

Quantiles are used because WPD can be highly skewed due to the cubic dependence on wind speed.

Extreme values may strongly affect mean and variance.

The interquartile structure provides a robust measure of the central spread of the energetic signal.

Thus:

FNRR = robust irregularity of WPD central distribution

This protects the descriptor from being dominated only by isolated extreme events.

## G.25 Usable energy formulation

The thesis distinguishes between:

$E_{free}$

and

$E_{usable}$

The usable energy is defined as:

$$E_{usable,z}(T) = [1 - FNRR_z(T)] \cdot E_{free,z}(T)$$

where:

| Symbol | Meaning |
|---|---|
| $E_{usable,z}(T)$ | Structurally usable energy for zone z and window T. |
| $E_{free,z}(T)$ | Physically available energy before irregularity adjustment. |
| $FNRR_z(T)$ | Regional non-regularity factor. |

This expression applies a structural penalty to the physically available energy.

## G.26 Boundedness of usable energy

Because:

$$0 \leq FNRR_z(T) < 1$$

then:

$$0 < 1 - FNRR_z(T) \leq 1$$

If:

$$E_{free,z}(T) \geq 0$$

then:

$0 \leq E_{usable,z}(T) \leq E_{free,z}(T)$

This proves that usable energy is bounded by free energy.

Therefore, the FNRR adjustment cannot create energy artificially.

It can only reduce the physically available energy according to structural irregularity.

## G.27 Monotonicity of usable energy with respect to FNRR

From:

$E_{usable} = (1 - FNRR) \cdot E_{free}$

assuming $E_{free}$ is fixed:

$\partial E_{usable}/\partial FNRR = -E_{free}$

Since:

$E_{free} \geq 0$

then:

$\partial E_{usable}/\partial FNRR \leq 0$

Thus, usable energy decreases monotonically as FNRR increases.

This is physically and methodologically coherent: higher irregularity reduces the structurally usable portion of energy.

### G.27.1 Additional mathematical properties

#### Scale behavior of FNRR

If all WPD values are multiplied by a positive constant $a > 0$, then:

$Y'_z(T) = a \cdot Y_z(T)$

and therefore:

$Q'_{25,z}(T) = a \cdot Q_{25,z}(T)$
$Q'_{75,z}(T) = a \cdot Q_{75,z}(T)$
$IQR'_z(T) = a \cdot IQR_z(T)$

If the numerical-stability term is scaled consistently as:

$\varepsilon'_z = a \cdot \varepsilon_z$

then:

$FNRR'_z(T) = FNRR_z(T)$

This shows that FNRR is invariant under positive scaling when the stability term is scaled consistently with the WPD unit.

This is important because FNRR measures relative structural irregularity, not absolute energetic magnitude.

## Sensitivity of usable energy to free energy

From:

$$E_usable = (1 - FNRR) \cdot E_free$$

assuming FNRR is fixed:

$$\partial E_usable/\partial E_free = 1 - FNRR$$

Since:

$$0 \leq FNRR < 1$$

then:

$$0 < \partial E_usable/\partial E_free \leq 1$$

This means that usable energy increases with free energy, but its rate of increase is reduced by regional irregularity.

## Interpretation of the FNRR penalty

The term:

$$1 - FNRR$$

acts as a structural usability factor.

When FNRR is low:

$$1 - FNRR \approx 1$$

most of the free energy remains structurally usable.

When FNRR is high:

$$1 - FNRR \to 0$$

the usable fraction decreases.

This behavior is mathematically coherent with the physical interpretation of regional irregularity.

## G.28 Interpretation of free and usable energy

The distinction can be summarized as:

$E_{free}$ → what the wind system physically offers
$E_{usable}$ → what remains after structural irregularity is considered

This distinction protects the thesis from overinterpreting gross energetic potential.

A zone with high free energy but high FNRR may not be as structurally favorable as a zone with moderate free energy and lower irregularity.

## G.29 FNRR over annual and quarterly windows

For annual analysis:

$FNRR_z(year) = FNRR_z(T_{year})$

For quarterly analysis:

$FNRR_z(quarter) = FNRR_z(T_{quarter})$

This allows regional irregularity to be evaluated at different temporal scales.

Annual FNRR supports long-term interpretation.

Quarterly FNRR supports subannual or seasonal interpretation.

Both are useful because wind-energy behavior may change across time scales.

## G.30 Projection toward 2028

The projection stage extends the historical analysis toward 2028.

For a projected window $T_{proj}$, the model estimates:

$\hat{W}PD_z(t), \quad t \in T_{proj}$

Then:

$\hat{E}_{free,z}(T_{proj}) = (\Delta t / 1000) \cdot \Sigma_{t \in T_{proj}} \hat{W}PD_z(t)$

and:

$FNRR_z(T_{proj}) = IQR(\hat{W}PD_z(T_{proj})) / [Q_{75}(\hat{W}PD_z(T_{proj})) + Q_{25}(\hat{W}PD_z(T_{proj})) + \varepsilon_z]$

Finally:

$\hat{E}_{usable,z}(T_{proj}) = [1 - FNRR_z(T_{proj})] \cdot \hat{E}_{free,z}(T_{proj})$

Projected values must be interpreted as model-based scenarios, not as direct observations.

## G.31 Uncertainty-aware energy projection

If prediction intervals are available for WPD, then energy can also be interpreted through lower and upper projected scenarios.

Let:

$L_{WPD,z}(t)$

and:

$U_{WPD,z}(t)$

be the lower and upper PI90 bounds for WPD.

Then:

$$E_{free,z}^{L}(T) = (\Delta t / 1000) \cdot \Sigma_{t \in T} L_{WPD,z}(t)$$

$$E_{free,z}^{U}(T) = (\Delta t / 1000) \cdot \Sigma_{t \in T} U_{WPD,z}(t)$$

This produces an energy interval:

$$E_{free,z}(T) \in [E_{free,z}^{L}(T), E_{free,z}^{U}(T)]$$

A corresponding usable-energy interval may be approximated as:

$$E_{usable,z}^{L}(T) = [1 - FNRR_z(T)] \cdot E_{free,z}^{L}(T)$$

$$E_{usable,z}^{U}(T) = [1 - FNRR_z(T)] \cdot E_{free,z}^{U}(T)$$

If FNRR uncertainty is also estimated, then interval propagation should include uncertainty in FNRR as well.

## G.32 Complete mathematical chain

The complete mathematical chain of the thesis can be summarized as:

$$x_z(t) \rightarrow \rho_z(t) \rightarrow WPD_z(t) \rightarrow \hat{W}PD_z(t+h) \rightarrow PI90_z(t+h)$$

then:

$$\hat{W}PD_z(T) \rightarrow E_{free,z}(T)$$

then:

$$WPD_z(T) \rightarrow FNRR_z(T)$$

finally:

$$E_{usable,z}(T) = [1 - FNRR_z(T)] \cdot E_{free,z}(T)$$

This chain connects observation, physics, prediction, uncertainty, irregularity, and energy.

## G.33 Scientific interpretation of the model

The model should be interpreted as a physical–statistical framework, not as a purely algorithmic system.

The physical component is represented by:

- wind-speed dynamics,
- air-density estimation,
- WPD construction,
- energy integration,
- and physical units.

The statistical component is represented by:

- distributional characterization,
- predictive modeling,
- performance metrics,
- persistence comparison,
- residual analysis,
- and uncertainty intervals.

The structural component is represented by:

- FNRR,
- free-energy adjustment,
- usable-energy estimation,
- and regional projection.

Together, these components form the doctoral contribution.

## G.34 Relationship with Chapter 2

Chapter 2 provides the physical and statistical characterization of wind speed.

It supports:

- descriptive statistics,
- Weibull fitting,
- Rayleigh comparison,
- ACF,
- PACF,

- FFT,
- Wavelet analysis,
- and zonal interpretation.

Mathematically, Chapter 2 establishes properties of:

$V_z = \{v_z(t)\}$

which later determine:

$WPD_z(t) = 0.5 \cdot \rho_z(t) \cdot v_z(t)^3$

Thus, Chapter 2 provides the empirical and physical basis for the energetic model.

## G.35 Relationship with Chapter 3

Chapter 3 provides the predictive modeling layer.

It estimates:

$\hat{W}PD_z(t+h)$

and evaluates model performance using:

- RMSE,
- MAE,
- $R^2$,
- Skill Score versus persistence,
- PI90 coverage,
- residual behavior,
- and horizon-based interpretation.

Mathematically, Chapter 3 defines the transition:

$WPD_z(t) \rightarrow \hat{W}PD_z(t+h)$

This transition is required before energy projection.

## G.36 Relationship with Chapter 4

Chapter 4 integrates prediction, uncertainty, FNRR, and energy.

It formalizes:

$E_{free,z}(T)$

$FNRR_z(T)$

$E_{usable,z}(T)$

and projection toward 2028.

Mathematically, Chapter 4 provides the closure:

prediction + uncertainty + irregularity → usable energy projection

This is the final scientific stage of the thesis.

## G.37 Relationship with repository results

This appendix supports the following repository folders:

04_RESULTS_COMPLETE/01_physical_characterization/
04_RESULTS_COMPLETE/02_model_comparison/
04_RESULTS_COMPLETE/03_pi90_uncertainty/
04_RESULTS_COMPLETE/04_fnrr_outputs/
04_RESULTS_COMPLETE/05_energy_projection/
04_RESULTS_COMPLETE/07_FIGURES/
04_RESULTS_COMPLETE/08_TABLES/

The appendix provides the mathematical explanation behind the evidence stored in those folders.

## G.38 Relationship with code

The formulation in this appendix is operationally connected to:

03_CODE/01_preprocessing/
03_CODE/02_physical_characterization/
03_CODE/03_classical_models/
03_CODE/04_machine_learning/
03_CODE/05_deep_learning/
03_CODE/06_hybrid_tdq/
03_CODE/07_energy_projection/
03_CODE/08_utils/

The folder 03_CODE/06_hybrid_tdq/ reflects the current repository naming. In this appendix, it should be interpreted as the hybrid physical–statistical integration stage of the doctoral workflow.

## G.39 Methodological assumptions

The formulation depends on the following assumptions:

1. Wind speed values used for WPD construction are physically admissible.
2. Air density is estimated through the density formulation implemented in the thesis workflow.

3. Units are harmonized before physical calculations.
4. WPD is nonnegative when wind speed and air density are nonnegative.
5. Prediction horizons are defined consistently.
6. Persistence is used as the main benchmark for predictive gain.
7. PI90 intervals are interpreted empirically.
8. FNRR is a dimensionless robust descriptor, not a universal constant.
9. Energy projection is model-based and should not be interpreted as direct observation.
10. Usable energy is bounded above by free energy.
11. The numerical-stability term $\varepsilon_z$ must be expressed in the same physical unit as WPD.

## G.40 Methodological cautions

This appendix does not claim that the wind system is deterministic.

It does not claim that FNRR eliminates uncertainty.

It does not claim that all projected values are guaranteed future values.

It does not claim that FNRR is a universal physical law.

Instead, it defines a physically consistent, mathematically bounded, and computationally traceable framework for interpreting wind-energy prediction under regional irregularity and uncertainty.

## G.41 Main propositions

### Proposition 1 — Nonnegativity of WPD

If:

$\rho \geq 0$

and:

$v \geq 0$

then:

$WPD = 0.5 \cdot \rho \cdot v^3 \geq 0$

Thus, WPD is physically nonnegative.

### Proposition 2 — WPD is dimensionally consistent

Given:

$\rho \rightarrow kg/m^3$
$v^3 \rightarrow m^3/s^3$

then:

$\rho v^3 \rightarrow kg/s^3 = W/m^2$

Thus, WPD has units of power per unit area.

### Proposition 3 — FNRR is dimensionless and bounded

For nonnegative WPD and $\varepsilon_z > 0$ with the same unit as WPD:

$FNRR_z(T) = [Q_{75,z}(T) - Q_{25,z}(T)] / [Q_{75,z}(T) + Q_{25,z}(T) + \varepsilon_z]$

The numerator and denominator have the same unit.

Therefore, FNRR is dimensionless.

Additionally:

$0 \leq FNRR_z(T) < 1$

A conservative practical statement is:

$0 \leq FNRR_z(T) \leq 1$

### Proposition 4 — Usable energy is bounded by free energy

If:

$E_{usable} = (1 - FNRR) \cdot E_{free}$

with:

$0 \leq FNRR < 1$

and:

$E_{free} \geq 0$

then:

$0 \leq E_{usable} \leq E_{free}$

Thus, usable energy cannot exceed free energy.

### Proposition 5 — Usable energy decreases with irregularity

If $E_{free}$ is fixed:

$\partial E_{usable}/\partial FNRR = -E_{free} \leq 0$

Thus, increasing irregularity decreases usable energy.

### Proposition 6 — FNRR measures relative irregularity

If WPD is scaled by a positive factor $a > 0$ and the stability term is scaled consistently:

$Y'_z(T) = a \cdot Y_z(T)$

$\varepsilon'_z = a \cdot \varepsilon_z$

then:

$FNRR'_z(T) = FNRR_z(T)$

Thus, FNRR measures relative structural irregularity rather than absolute energetic magnitude.

## G.42 Final synthesis

The doctoral model can be summarized as:

Meteorological system:
$x_z(t)$

Physical transformation:
$WPD_z(t) = 0.5 \cdot \rho_z(t) \cdot v_z(t)^3$

Prediction:
$\hat{W}PD_z(t+h) = f_m(\Phi_z(t); \theta_m)$

Uncertainty:
$PI90_{z,m,h}(t) = [L_{z,m,h}(t), U_{z,m,h}(t)]$

Free energy:
$E_{free,z}(T) = (\Delta t / 1000) \cdot \Sigma\, WPD_z(t)$

Regional irregularity:
$FNRR_z(T) = [Q_{75,z}(T) - Q_{25,z}(T)] / [Q_{75,z}(T) + Q_{25,z}(T) + \varepsilon_z]$

Usable energy:
$E_{usable,z}(T) = [1 - FNRR_z(T)] \cdot E_{free,z}(T)$

This sequence is the mathematical backbone of the doctoral thesis.

## G.43 Final statement

The physical and mathematical formulation of the thesis is built on one central idea:

Wind-energy potential cannot be interpreted only from wind speed;
it must be constructed physically, predicted statistically,
bounded probabilistically, corrected structurally,
and interpreted energetically.

The role of FNRR is to introduce a mathematically bounded, dimensionally consistent, and robust structural descriptor that prevents the overinterpretation of free energy as fully usable energy.

For that reason, this appendix provides the formal support for the final doctoral contribution:

a physical–statistical framework for forecasting and interpreting
regional wind-energy potential under uncertainty and structural irregularity.